Special Instructions: You must write your name and netid on every sheet of the exam.

ECE 340 – Spring 2026

Name_____

Net ID#_____

Section_____

ECE 340 – EXAM 1
Thursday March 5th, 2026 7:00 – 8:00 p.m.

1002 ECEB A (He), C (Bogdanov), Sections X (Bayram)

NOTE: This is a closed book and closed notes exam. The exam consists of three problems. Unless stated otherwise, do your work on the page of the problem and if necessary on the preceding blank page. A simple scientific calculator is allowed but it must have no wireless or internet connectivity. Cell phones are not permitted.
Be sure to explicitly show the units in your work, as well as in your answers. Circle your answer. Be neat!

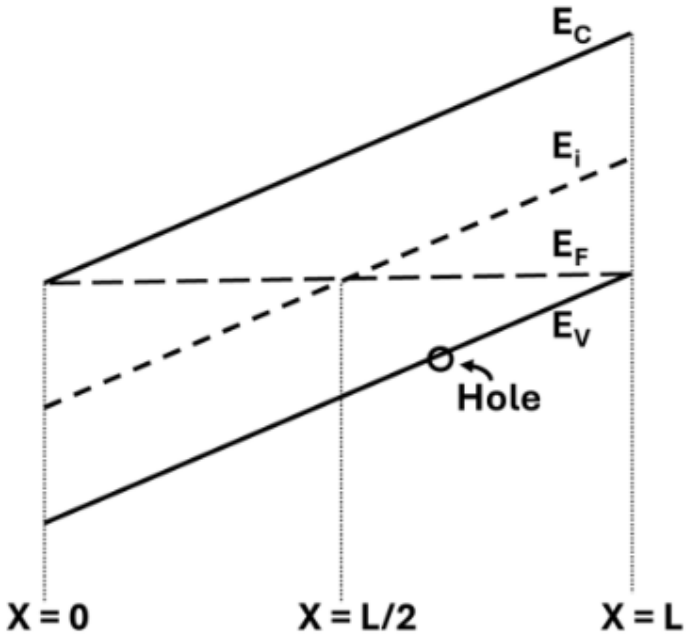
If we cannot read or follow your work, you get zero credit!

For each problem, you must show complete work and **indicate your reasoning**. No credit will be given if you do not show the complete work *and describe your procedure*, even if the answer is correct. Write your name, ID#, and section on this page and **sign below**.

	Prob.#1		Prob.#2		Prob.#3	
	33 pts.		33 pts.		34 pts.	
a	8		10		8	
b	10		13		7	
c	6		10		7	
d	9		---		12	
e	---		---		---	
f	---		---		---	
g	---		---		---	
T	33		33		34	

Signature:_____ Your Exam Score:_____

Problem 1



A semiconductor under equilibrium conditions is characterized by the energy band diagram given above. Bandgap of the semiconductor (E_G) is 1.2 eV, $n_i = 10^{11}$ /cm^3, kT = 0.026 eV.

a) At X = L/2, what are the i) electron concentration n(X) for X = L/2 and ii) hole concentration p(X) for X = L/2? [8 points]

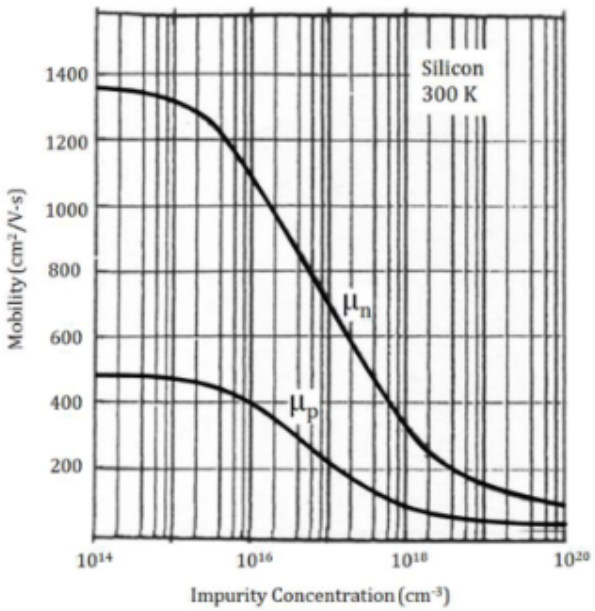
b) At X = L/4, i) is the material n-type or p-type? ii) what is the electron concentration n(X=L/4)? [10 points]

c) What is the approximate kinetic energy of the hole shown in the diagram? [6 points]

d) If L = 10 μm, what is the magnitude of the electric field inside the semiconductor? [9 points]

Problem 2

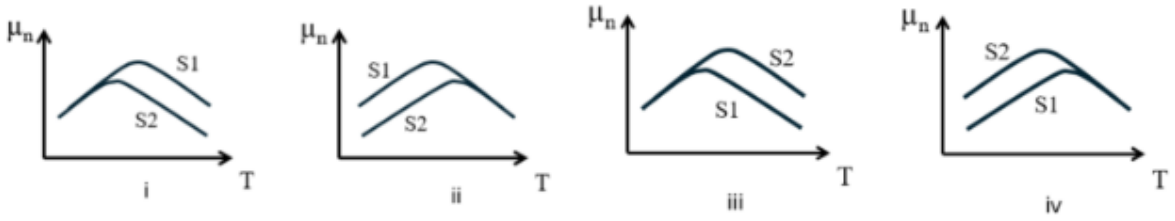
Consider two silicon samples at temperature 300K. Sample one (**S1**) is doped with acceptors with a concentration $N_a = 1 \times 10^{17}$ cm^{-3}. Sample two (**S2**) is doped both with acceptors having a concentration $N_a = 1 \times 10^{17}$ cm^{-3} and with donors at a concentration $N_d = 5 \times 10^{17}$ cm^{-3}. The electron and hole mobility *vs.* impurity concentration is shown in the figure below. The intrinsic carrier concentration for silicon n_i at 300K is 1.5×10^{10} cm^{-3}.



a) Calculate the conductivity σ $(\Omega\text{-cm})^{-1}$ of sample **S1**. [10 points]

b) Calculate the conductivity σ $(\Omega\text{-cm})^{-1}$ of sample **S2**. Specify the doping concentration, the value of majority carrier mobility, and the impurity concentration you used to determine the mobility value. [13 points]

c) Consider figure *i)* through figure *vi)* as shown below. Which figure correctly illustrates the **electron mobility** *vs.* temperature relation for sample S1 and sample S2? Briefly justify your answer. (10 points)



Problem 3

A uniformly doped n-type silicon sample at T = 300K has the following properties:

Doping concentration	$N_D = 1.5 \times 10^{16}$ cm^{-3}
Carrier lifetimes	$\tau_n = \tau_p = 15$ μs
Electron mobility	$\mu_n = 1350$ cm$^2 \cdot$V$^{-1} \cdot$s^{-1}
Hole mobility	$\mu_p = 450$ cm$^2 \cdot$V$^{-1} \cdot$s^{-1}
Intrinsic concentration	$n_i = 1.5 \cdot 10^{10}$ cm^{-3}

A 10% numerical accuracy is acceptable in this problem.

a) The sample is illuminated uniformly with a constant optical generation rate $G = 10^{19}$ cm^{-3}s^{-1}. Calculate the steady-state excess electron and hole concentrations δn and δp. Is this a low-level or high-level injection? Justify your answer briefly. [8 points]

b) The illumination is suddenly turned off at $t = 0$. Write the expression for the total hole concentration for all $t > 0$ as a function of time. Plugging in the numerical values is optional. [7 points]

c) For the steady-state condition of question a), calculate the quasi-Fermi levels for holes relative to the intrinsic Fermi level (F_p - E_i). Express your answers in meV. [7 points]

d) The illumination is changed such that at $t = 0$ a gradient in the excess carrier concentration is produced that follows a linear profile from $x = 0$ to $x = 1\ \mu m$. Assuming $\delta n(x = 0) = \delta p(x = 0) = 10^{14} cm^{-3}$ and $\delta n(x = 1\ \mu m) = \delta p(x = 1\ \mu m) = 10^{16} cm^{-3}$ and that there is no electric field, calculate the electron and hole diffusion current densities at $x = 0.5\ \mu m$. [12 points]